



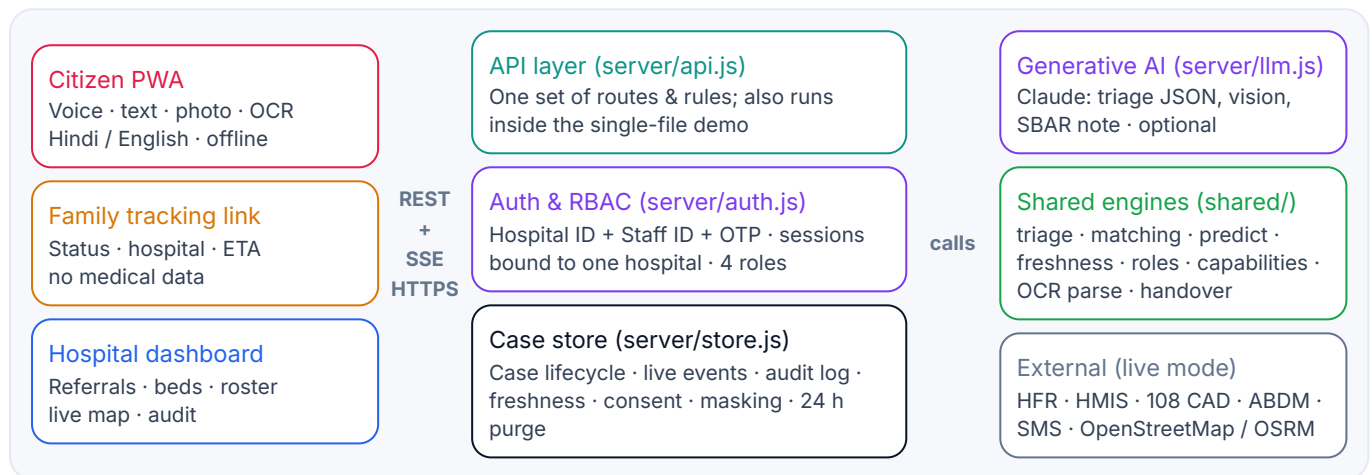
Technology Architecture & Technical Approach

How Medreach is built: components, data flow, AI design, security, scalability and the seven technologies in code.

☆ MPOne Idea & Innovation Hackathon 2026 🏠 Challenge 5 · AI for Public Services · Healthcare

👤 Team Leader: Prakhar Porwal 🌐 medreach-wtne.onrender.com 📄 github.com/prakharporwal518-collab/Medreach

1 System architecture



The shared/ engines run on the server and on the phone, so triage and matching still work offline.

2 Technology stack

Layer	Choice and reason
Frontend	Progressive Web App (HTML, CSS, JS modules): installable, fast on low-end phones, offline via service worker.
Backend	Node.js 22 + Express 5: a thin adapter over framework-independent API code; Server-Sent Events for live updates.
AI	Anthropic Claude via official SDK: structured JSON output, vision for photos, SBAR drafting; server-side fallback; strict timeouts.
Maps	Leaflet + free OpenStreetMap tiles (no API keys) · OSRM road routing with straight-line fallback · live markers.
OCR / Voice	Tesseract.js on the device (card never uploaded) · Web Speech API for Hindi / English speech in and out.
Hosting	Render cloud with health check and auto-deploy on every merge.

3 Data flow of one emergency



Describe → /api/triage (AI + rules) → /api/match (deterministic ranking) → /api/cases (case token + family token) → /request → hospital SSE → staff /respond (accept + bay) → /transport → live position events → /arrived → retention purge after 24 h.

4 The seven technologies in code

Technology	Technical approach	Code
Generative AI	Claude returns schema-validated JSON (type, severity, red flags, capabilities) and drafts the SBAR note. Refusals and timeouts fall back to rules.	server/llm.js
Voice Bots	Speech recognition (hi-IN / en-IN) for input; speech synthesis reads acceptance, safety steps and directions; console announces referrals.	public/js/voice.js
Computer Vision	Claude vision describes visible findings; offline pixel analysis flags likely blood or burns. Findings can only raise severity.	server/llm.js, public/js/vision.js
OCR	Tesseract.js reads Ayushman / ABHA cards and prescriptions on the phone; a parser extracts name, age, IDs, allergies.	public/js/ocr.js, shared/ocr-parse.js
Predictive Analytics	Hour-of-day traffic ETA; Poisson probability a bed is still free on arrival, weighted by data freshness; ER wait; 8-hour arrival forecast.	shared/predict.js
Conversational AI	Asks one follow-up question at a time (conscious? breathing normally? heavy bleeding? age?) and updates severity after each answer.	shared/triage.js
Mobile Apps	Installable PWA, bilingual UI, service-worker cache of app shell and engines, network-first API with cached fallback.	public/, public/sw.js

5 AI design principles



AI understands, rules decide

AI only extracts requirements. The hospital choice is made by a deterministic, testable engine, and every step is explained in the app.



Safety floor

Severity = max(AI, rules). The AI can never downgrade a red-flag case. Offline or on error, the rules engine is used.



Honest language

"This may indicate...", "Reported capacity – not a guarantee of admission", "Simulated demo data".

6 Security & privacy architecture

Control	Implementation	Control	Implementation
Staff login	Hospital ID + Staff ID + OTP; session bound to one hospital	Consent	Patient details shared only if the family agrees
Authorisation	Role-based: nodal officer, desk, resources, state admin	Masking	ABHA / Ayushman IDs masked; pick-up point rounded ~100 m
Case access	Case token for owner; read-only track token for family	Citizen data	Profile & records stay on the device (PBKDF2 password hash)
Audit	Every login, change, decision and denied attempt logged	Retention	Personal / health details deleted 24 h after hand-over

7 Scalability & reliability

- Scale out: stateless API code; move the in-memory store to PostgreSQL and live events to Redis pub/sub to run many instances.
- Graceful degradation: AI → rules engine · road routing → straight line · map tiles → plain grid with pins · network → offline PWA.
- Quality: 51 automated tests (triage, matching, scoring, auth, roles, privacy) plus browser checks.